Procedural Istanbul

Estimating corridor demand from live speeds, and an open SUMO digital twin of the Maslak – 15 July Bridge corridor

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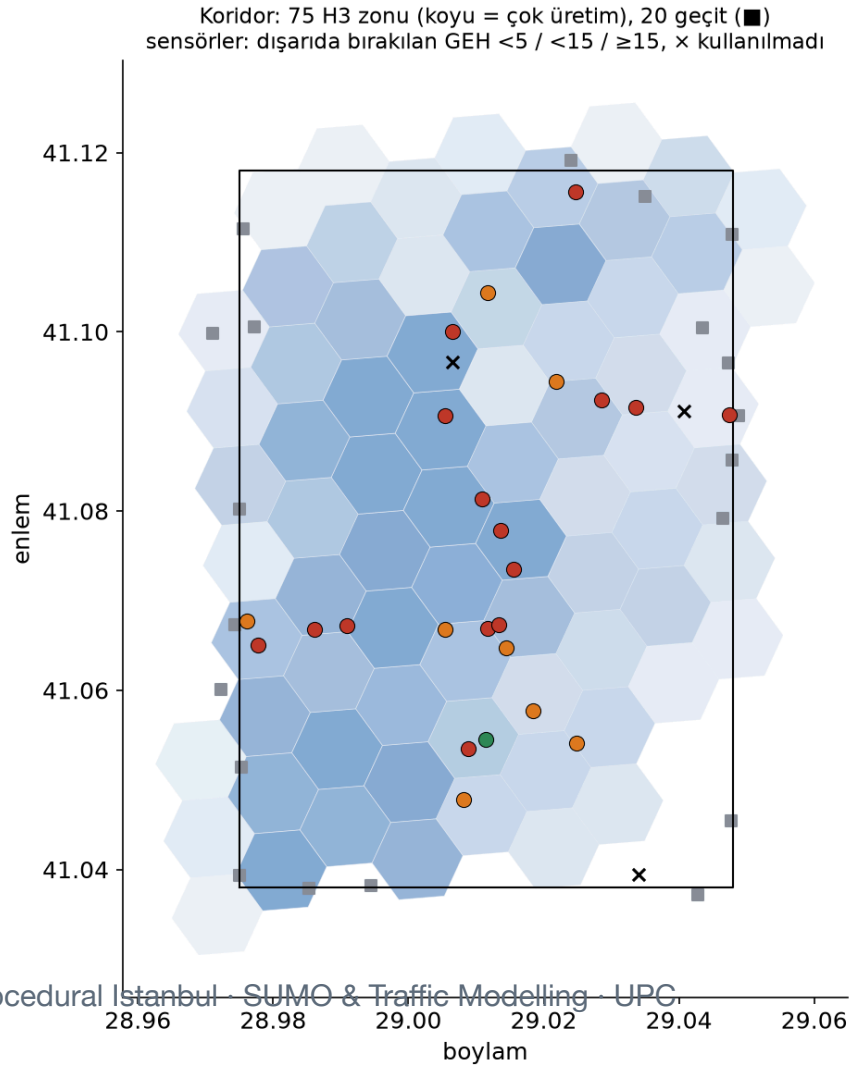
github.com/aaasiroglu/procedural-istanbul · MIT

Motivation

- A simulator needs **demand**: who travels from where to where (an OD matrix).
- Istanbul's traffic centre (İBB UYM) publishes **speeds** for ~12,700 road segments every minute, but **no flows**.
- Real counts are sparse: 27 sensors in a 52 km² corridor; 9,025 OD pairs to estimate. The inverse problem is **ill-posed**.
- Common practice turns speeds into pseudo-counts, injecting a certainty the data does not have, and rarely reports which part of the estimate comes from data and which from assumption.

Goal: an open, reproducible pipeline from public data to a calibrated SUMO scenario of the corridor, with honest uncertainty and validation on held-out counts.

Study area and data



1 · Building the SUMO network

OSM → SUMO with `netconvert`

```
netconvert --osm-files koridor.osm.xml --type-files osmNetconvert.typ.xml \
  --keep-edges.by-vclass passenger,bus --keep-edges.components 1 --remove-edges.isolated \
  --geometry.remove --edges.join --roundabouts.guess --junctions.join \
  --osm.turn-lanes --osm.lane-access \
  --tls.guess-signals --tls.join --tls.default-type static --tls.cycle.time 90 --proj.utm
```

Result	
Edges / junctions	21,174 / 8,963
Signalised junctions / programs	293 / 179
Car edges mutually reachable (SCC)	98.3 %
Metrobüs (bus-only carriageway)	11.8 km
15 July Bridge	3 lanes per direction

Lesson: <code>--tls.discard-simple</code> removed ~2/3 of the corridor's signals (mid-block pedestrian lights, split junctions of divided roads), overstating arterial capacity. Left out on purpose.

Network checks before any demand

Smoke test (`randomTrips.py` + `sumo`)

Load	Collisions	Teleports
3,000 veh/h	0	1.7 %

Teleports concentrate at two unsignalised junction clusters (yield deadlocks): Harbiye and Dikilitaş–Fulya.

Live data on the network

- İBB segments matched to SUMO edges by heading-aware sampling: **95.5 %** of the corridor's segments.
- Segment geometry from `Segments/0` (13,534); `Segments/1` silently omits motorways.
- Observed share of road length: motorway 99 %, trunk 100 %, primary 94 %, secondary 92 %, residential 8 %.

İBB's calibrated 2020 model on the SUMO network

- İBB's 29,663 road links (KMZ) mapped onto SUMO edges, **direction-aware**: İBB draws each carriageway of a divided road as its own one-way link, a few metres off OSM, so the nearest link is often the opposite one.
- Matched: motorway 92 %, trunk 80 %, primary/secondary 77–78 %.
- Carried over: capacity per direction, free-flow speed, **calibrated BPR**:

$$t = t_0 \left(1 + \alpha \left(\frac{v}{c}\right)^{\beta}\right) \qquad \text{expressways: } \alpha = 0.158,\ \beta = 3.483 \qquad \text{streets: } \alpha = 0.924,\ \beta = 3.464$$

- 540-zone OD matrix: its zone order cannot be recovered from the published files. A distance–flow correlation test gave r = –0.08 for the candidate order, no better than random orders, so the matrix is **not used**.

2 · Demand from speeds

Speeds as censored observations

On the triangular fundamental diagram (Newell 1993; Daganzo 1994):

- **Free-flow speed says nothing about flow** (anything from 0 to capacity): an **upper bound**.
- **Congested branch:** q = w·k_j·v / (v + w). A queue means demand ≥ throughput: a **lower bound**.
- **Branch selection:** decided by İBB's own congestion class, which carries the operator's knowledge of each segment.

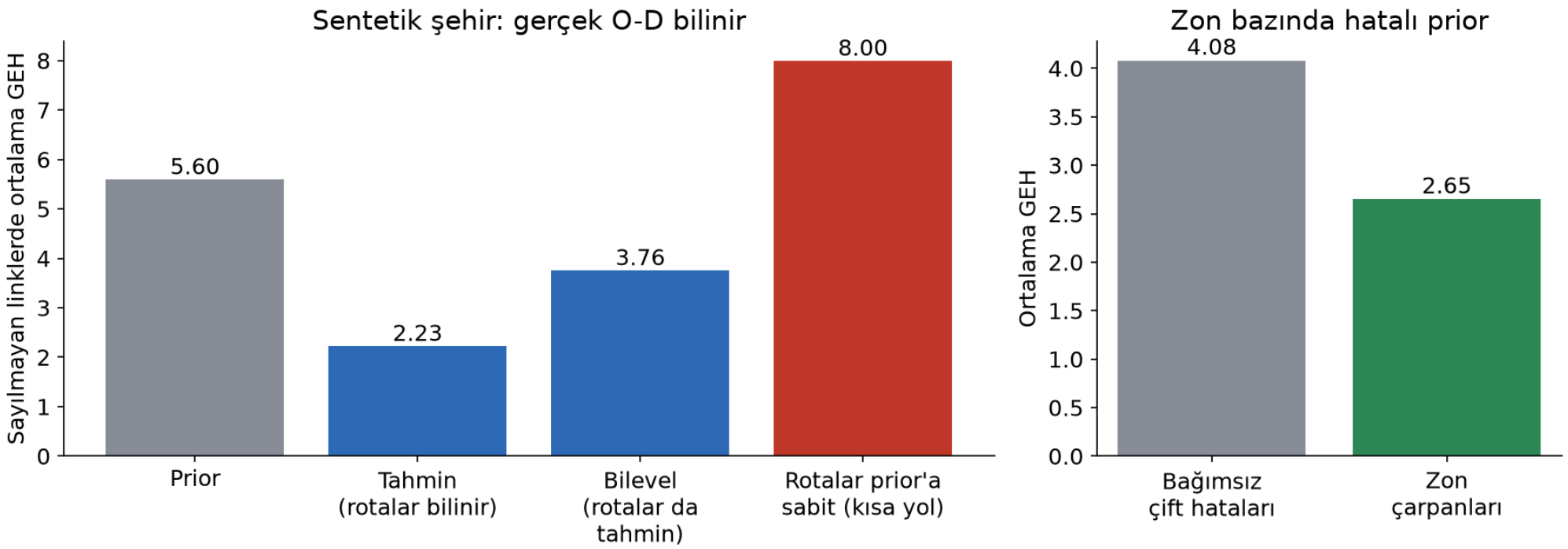
$$\min_{x \geq 0} \sum_{\text{counts}} \frac{((Px)_e - y_e)^2}{s_e^2} + \sum_{\text{lower}} \frac{[y_e - (Px)_e]_+^2}{s_e^2} + \sum_{\text{upper}} \frac{[(Px)_e - y_e]_+^2}{s_e^2} + \mathrm{KL}_\lambda(x \,\|\, x_0)$$

Validated on real counts: the rule holds only for uninterrupted flow. On signalised streets low speed is also red time; those bounds inflated demand by a third. Lower bounds are now kept for motorways and trunk roads only.

Estimator design

- **Bilevel:** UE assignment (Frank–Wolfe, BPR) and estimation alternate; the route choice is averaged over rounds (MSA) to damp oscillation (Spiess 1990; Yang et al. 1992).
- **Correlated prior errors:** x_{ij} = x_{0,ij} · exp(a_i + b_j + u_{ij}). With independent per-pair errors, the prior is near certain about any link flow, which sums many pairs; zone production/attraction factors let a count correct a zone's whole traffic.
- **Uncertainty:** Laplace standard errors via Woodbury (an m × m solve rather than n × n), and a **data share** per OD pair, 1 – σ²_{post} / σ²_{prior}.
- **Numerics:** optimisation in log space (z = x₀·e^u) keeps L-BFGS well conditioned when prior values span 12 orders of magnitude.
- **Validation:** GEH on each count **left out in turn**, never only on the counts that were fitted.

Synthetic benchmark (true OD known)



- Flows on uncounted links: GEH 5.6 → **2.2** (known routes), **3.8** (routes estimated too); freezing routes at the prior gives 8.0.
- 96 % of true values lie within 2.5 standard errors.
- Individual OD cells barely improve (36 counts, 90 pairs): **flows are identifiable, cells are not**, and the data share reports exactly that.

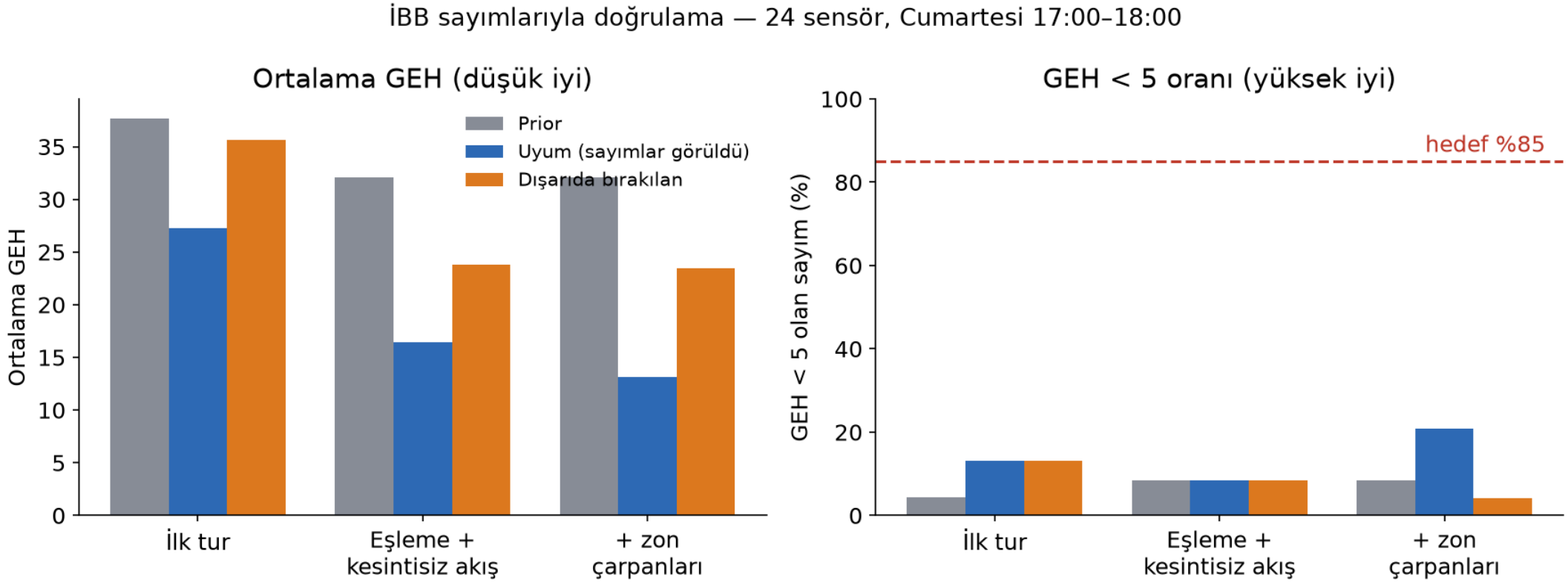
3 · Validation and the SUMO run

Counts on the SUMO network

- 25 of 27 İBB sensors placed as **screenlines** square to the road: both directions, all lanes.
- **Daily → hourly:** probe-vehicle share of the hour in the sensor's geohash cell, the same day (Saturdays of Oct 2024, 17:00–18:00; 3.6–5.1 % of the day).
- **The audit mattered.** Expressway sensors had snapped to side streets:

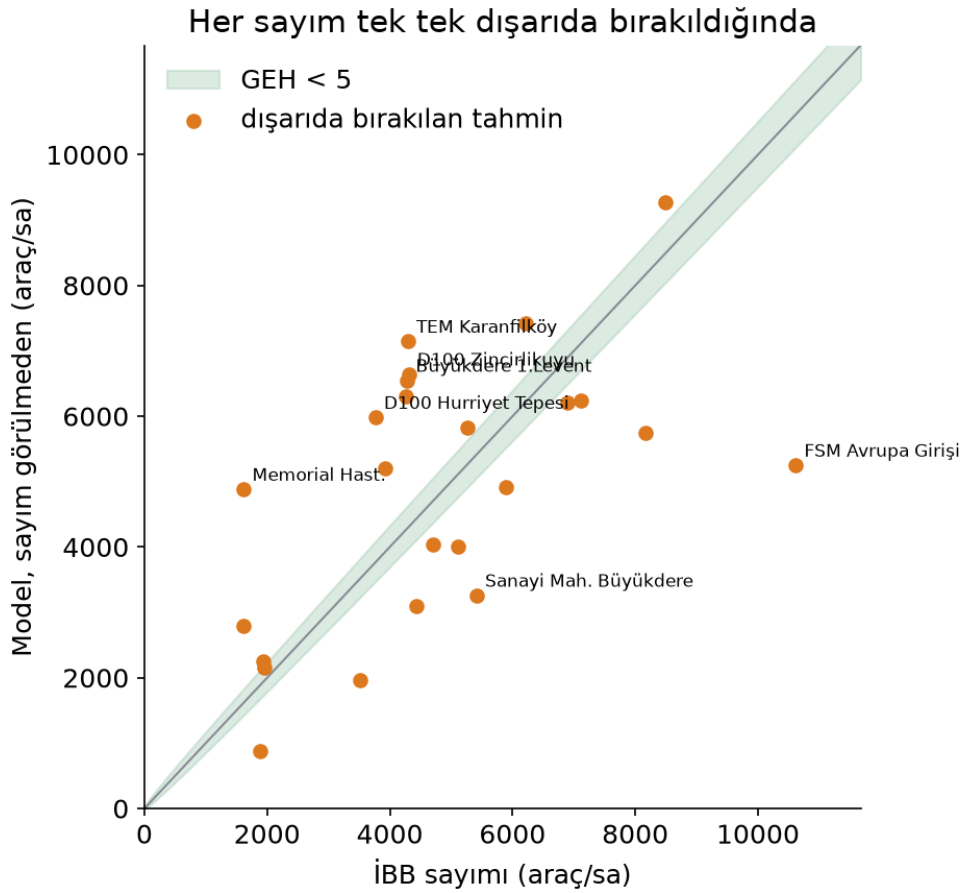
Sensor	Daily	First placement	Fixed
Mecidiyeköy	110 k	parallel street (no route)	O-1
Çağlayan junction	103 k	one-lane street	O-1
TEM Karanfilköy	103 k	Uğur Mumcu Cd.	O-2
FSM toll plaza	81 k	same edges as bridge entrance (224 k)	dropped

Held-out GEH: progress, not yet success



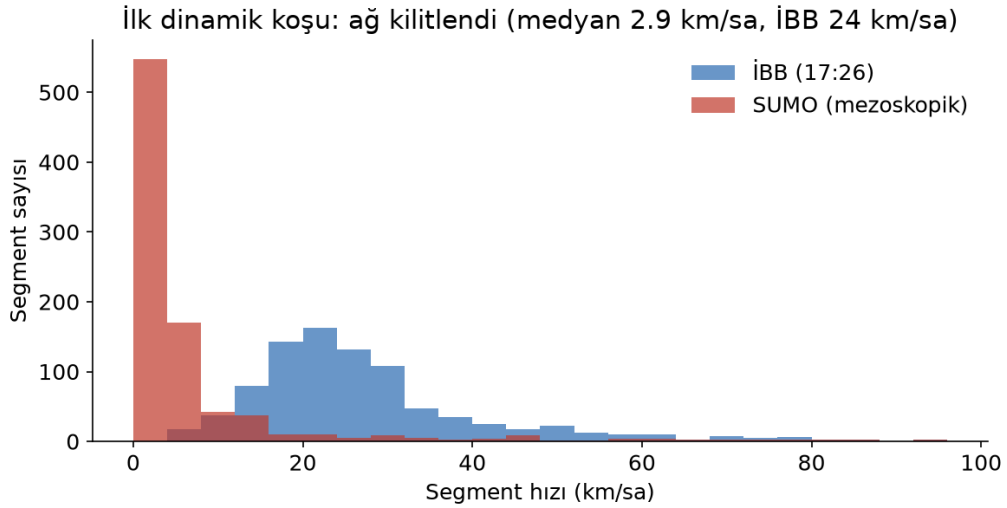
Mean held-out GEH 35.6 → 23.4. With İBB's calibrated capacities and BPR: fit 12.2, held-out 24.0, unchanged. **The GEH < 5 target (85 %) is not met.**

Where the remaining error comes from



1. **Count definitions conflict.** Three O-D sensors sit on the same modelled flow, yet read 4 291 / 7 108 /

Dynamic test in SUMO




```
od2trips --taz-files taz.xml \
  --tazrelation-files od.xml --spread.uniform
sumo --mesosim --meso-junction-control.limited \
  --meso-tls-flow-penalty 1 \
  --device.rerouting.probability 1 \
  --device.rerouting.period 120
```

- 128.6 k veh/h (estimate before counts), 90 min.

4 · The 3D twin

Procedural Istanbul in the browser



Suspension bridge (towers, cables, hangers) and Ortaköy Mosque



Limitations

- Held-out validation target not met yet (GEH < 5 for 4 % of counts against a target of 85 %).
- One time slice (a Saturday, 17:00); day types and hours not yet modelled.
- Static assignment inside the estimator: no turn bans and no spillback; the dynamic run is not yet consistent with the estimated demand.
- Prior parameters chosen by hand (storeys, trip rates, gateway shares); all listed with every estimate.
- The novelty of "speeds as censored observations" and "data share" still has to be confirmed by a systematic literature review.

Next steps

1. **Counts:** confirm sensor direction and lane coverage with İBB; resolve the conflicting O-2 sensors.
2. **Prior:** an observed PM-peak car OD from the 2012 household survey. Neighbourhoods are placed by their muhtarlık offices (963 points); external origins go to gateways by shortest path on İBB's 2020 network.
3. **SUMO:** scaled runs (25 / 50 / 75 / 100 %) of the count-anchored demand, junction fixes, and microscopic runs for the bridge approaches.
4. **Time:** hourly density archive since 2020 for day-type and hour profiles; continuous archiving of live speeds.
5. **Applications:** departure-time, mode and bridge advice for drivers; bottleneck ranking and scenario comparison for the city, back-tested on archived data.

Reproducibility

```
npm run sumo:net -- --area=koridor          # OSM → SUMO network + report
npm run sumo:map-ibb -- --area=koridor      # İBB live segments → edges
npm run sumo:smoke -- --area=koridor        # random-traffic network test
uv run --group sumo python tools/sumo/map_ibb_model.py --area=koridor   # İBB 2020 capacity & BPR
uv run --group sumo python tools/od/counts.py --area=koridor --month=2024-10 --weekday=5 --hour=17
uv run --group sumo python tools/od/corridor.py --area=koridor --speeds … --counts … --tag counts
uv run --group sumo python tools/sumo/run_od.py --od-dir data/od/koridor/<run>   # od2trips + SUMO
npm run test:od                             # synthetic benchmark, 28 checks
```

Every assumption, intermediate result and negative finding is recorded with dates and numbers in `tools/*/README.md`.

References

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Thank you

Questions and feedback welcome

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